

Dimensional Reduction in Supergravity and Duality Symmetries

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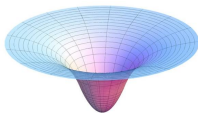
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Introduction

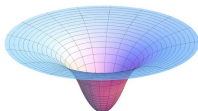
Gravity

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Super symmetry

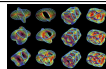
a symmetry mixing the degrees of freedom of fermions and bosons.

It appears in

Supergravity

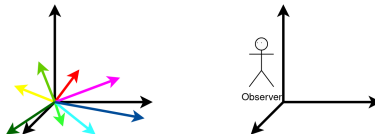


String Theory



!!Both theories predict extra dimensions!!

We need a way to describe how a 4-dimensional observer would see a 4 + E-dimensional universe.



$$S_{4+E} = \int dx^{4+E} \mathcal{L}_{4+E} \longrightarrow S_4 = \int dx^4 \mathcal{L}_4$$

This procedure is called **dimensional reduction**. Once we've reduced the action, we can also look for hidden symmetries it develops in its reduced form symmetries that were invisible in 4 + E dimensions.

4 + E gravity description

A warm-up: electromagnetism

Maxwell equations

- $\vec{\nabla} \cdot \vec{E} = \mu_0 c^2 \rho,$
- $\vec{\nabla} \times \vec{B} = \mu_0 \vec{j} + \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t},$
- $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t},$
- $\vec{\nabla} \cdot \vec{B} = 0.$

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Covariant formulation

- Four-current: $J^\mu = (c\rho, \vec{j}),$
- Four-potential: $A^\mu = (V/c, \vec{A}),$
- Field strength: $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$

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All four equations are packaged into a single action, the Maxwell action in vacuum:

$$S_F = \int dx^4 \frac{1}{4} F_{\mu\nu} F^{\mu\nu}$$

4-dimensional gravity

- General relativity encodes spacetime deformation in a metric:

$$g_{\mu\nu} = \begin{pmatrix} g_{00} & g_{01} & g_{02} & g_{03} \\ g_{01} & g_{11} & g_{12} & g_{13} \\ g_{02} & g_{12} & g_{22} & g_{23} \\ g_{03} & g_{13} & g_{23} & g_{33} \end{pmatrix}$$

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Einstein's equation: $R_{\mu\nu} + \frac{1}{2}g_{\mu\nu}R = 0$.

Going to 4 + E dimensions

- We enlarge the metric:

$$\hat{g}_{\hat{\mu}\hat{\nu}} = \begin{pmatrix} 4 \times 4 & 4 \times E \\ E \times 4 & E \times E \end{pmatrix}$$

A little hat mean that we are in high dimension.

- We assume no field depends on the internal coordinates.
- The action is rewritten accordingly:

$$S_{\hat{g}} = \int dx^4 \int \frac{dy^E}{\rho(E)} \sqrt{-\hat{g}} \hat{R}$$

Reduction

We make an ansatz on the metric,

$$\hat{g}_{\hat{\mu}\hat{\nu}} = \begin{pmatrix} g_{\mu\nu} - A_{\mu}^{(1)\gamma} A_{\nu\lambda}^{(1)} & -A_{\mu\beta}^{(1)} \\ -A_{\nu\alpha}^{(1)} & -h_{\alpha\beta} \end{pmatrix},$$

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and after some calculation, the reduced action is:

$$S_{\hat{g}} = \int dx^4 \sqrt{-g} \left(R + \frac{1}{4} F^{\mu\nu\alpha} F_{\alpha\mu\nu} - \frac{1}{4} g^{\rho\lambda} \partial_{\lambda} h_{\alpha\beta} \partial_{\rho} h^{\alpha\beta} \right).$$

Low dimension gravity, E electromagnetism field

$O(E, E)$ symmetry

Same idea, string theory version: the NS-NS action

In string theory we don't use the pure Einstein-Hilbert action, but a variant of it, the NS-NS action:

$$S = \int dx^4 \int \frac{dy^E}{\rho(E)} \sqrt{-\hat{g}} e^{-\hat{\phi}} \left(\hat{R} + \partial_{\hat{\mu}} \hat{\phi} \partial^{\hat{\mu}} \hat{\phi} - \frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho} \right),$$

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The two-form $B_{\hat{\mu}\hat{\nu}}$:

- Field strength:
 $H_{\mu\nu\rho} = \partial_{\mu} B_{\nu\rho} + \text{cyclic perms.}$
- Parametrized via the vielbein:

$$T^{\mu} = V^{\mu}_{\hat{r}} \hat{V}^{\hat{r}}_{\hat{\mu}} \hat{T}^{\hat{\mu}}, \quad T^{\alpha} = V^{\alpha}_{\hat{a}} \hat{V}^{\hat{a}}_{\hat{\mu}} \hat{T}^{\hat{\mu}},$$

$$T_{\mu} = V^{\hat{r}}_{\mu} \hat{V}^{\hat{\mu}}_{\hat{r}} \hat{T}_{\hat{\mu}}, \quad T_{\alpha} = V^{\hat{a}}_{\alpha} \hat{V}^{\hat{\mu}}_{\hat{a}} \hat{T}_{\hat{\mu}}.$$

The dilaton $\hat{\phi}$:

- Shifted dilaton:
 $\phi = \hat{\phi} - \frac{1}{2} \log h$
- $\partial_{\alpha} \hat{\phi} = 0$

The reduced action

Carrying out the reduction, we obtain:

$$S = \int dx^4 Ve^{-\phi} \left(\textcolor{red}{R} + \textcolor{blue}{\partial}_\mu \phi \textcolor{blue}{\partial}^\mu \phi + \frac{1}{8} \text{tr}(\partial_\mu M^{-1} \partial^\mu M) \right. \\ \left. - \frac{1}{4} \mathcal{F}_{\mu\nu}^i (M^{-1})_{ij} \mathcal{F}^{\mu\nu j} - \frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho} \right),$$

Low dimension gravity, E modified electromagnetism field, Shifted dilaton field, low dimension two form field

with M built from the internal metric h and the two-form B:

$$M = \begin{pmatrix} h^{-1} & -h^{-1}B \\ Bh^{-1} & h - Bh^{-1}B \end{pmatrix}.$$

O(E, E) invariance

We define the action of the O(E, E) group as

$$A^i \rightarrow \Omega_j^i A^j,$$

$$M \rightarrow \Omega^T M \Omega,$$

where $\Omega \in O(E, E)$.

...the action is left completely invariant!

Conclusion

Summary and outlook

■ Physical interpretation:

- Coupling between gravity and electromagnetism in low dimensions can emerge from pure gravity in higher dimensions.
- The NS-NS action is $O(E, E)$ invariant a genuinely stringy symmetry known as T-duality.

■ Prospects:

- searching for other hidden symmetries,
- allowing the fields to depend on the internal coordinates.

Definition of $O(E, E)$

A square $E \times E$ matrix M belongs to $O(E, E)$ if, for the matrix

$$\zeta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

which can be used as a metric for the group $O(E, E)$ since ζ has E negative and E positive eigenvalues in a basis rotated from the diagonal one M satisfies

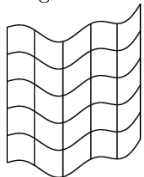
$$M^T \zeta M = \zeta.$$

Vielbein formalism

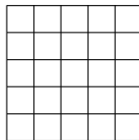
A new basis

$$g_{\mu\nu} V_a^\mu V_b^\nu = \eta_{ab},$$

Original Basis



New Basis



Vielbein
map

Curvature in this basis

$$\hat{R}_{\hat{\mu}\hat{\nu}\hat{\rho}\hat{\sigma}} = \partial_{\hat{\mu}} \hat{\omega}_{\hat{\nu}\hat{\rho}\hat{\sigma}} + \hat{\omega}_{\hat{\mu}\hat{\rho}}^{\hat{\tau}} \omega_{\hat{\nu}\hat{\tau}\hat{\sigma}} - (\mu \leftrightarrow \nu),$$

where the ω 's are defined by

$$\hat{\omega}_{\text{trs}} = -\hat{\Omega}_{\hat{\tau}\hat{\rho}\hat{\sigma}} + \hat{\Omega}_{\hat{\rho}\hat{\sigma}\hat{\tau}} - \hat{\Omega}_{\hat{\sigma}\hat{\tau}\hat{\rho}},$$

and the Ω 's in turn come from the vielbein:

$$\hat{\Omega}_{\hat{\rho}\hat{\sigma}\hat{\tau}} = \frac{1}{2} (\hat{V}_{\hat{\rho}}^{\hat{\mu}} \hat{V}_{\hat{\sigma}}^{\hat{\nu}} - \hat{V}_{\hat{\rho}}^{\hat{\nu}} \hat{V}_{\hat{\sigma}}^{\hat{\mu}}) \partial_{\hat{\nu}} \hat{V}_{\hat{\mu}\hat{\tau}},$$